

MEDREACH

The right hospital, right now.

Emergency coordination that connects a citizen, a capable hospital and ambulance response before the patient arrives.

CITIZEN

Speak or type in Hindi / English

HOSPITAL

Accept referral + prepare ER

108 / 102

Dispatch + live status

Hindi voice AI

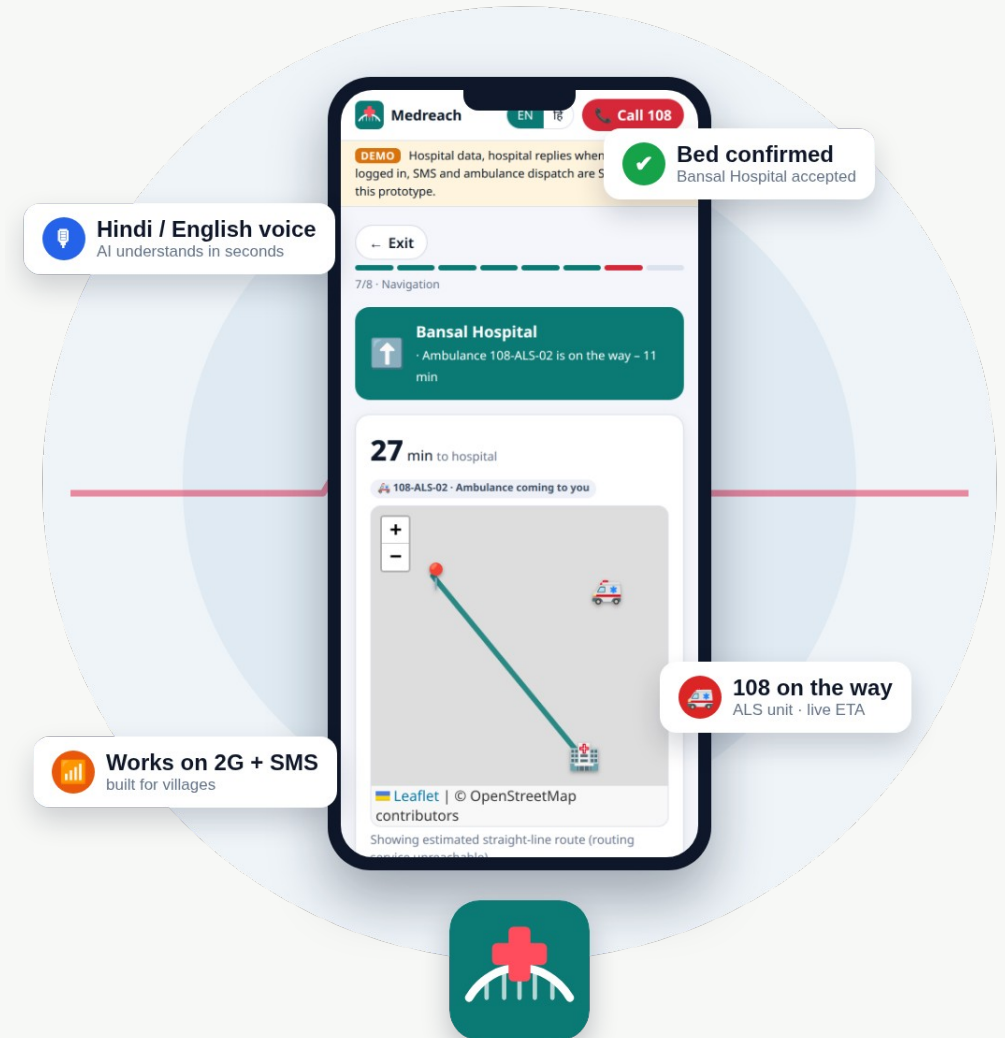
Accept before travel

108 / 102 linked

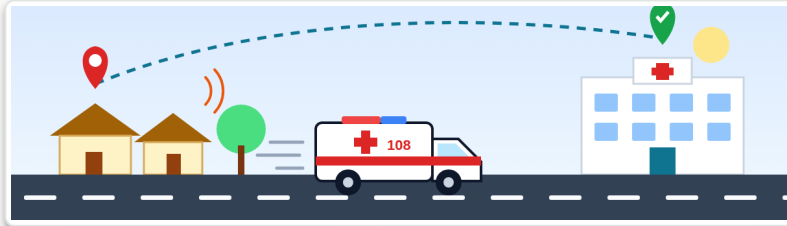
2G · SMS · offline

Working prototype: 16 hospitals · 4 districts · 9-ambulance practice 108 fleet · Hindi + English

Team MedReach | Prototype: medreach-2.onrender.com

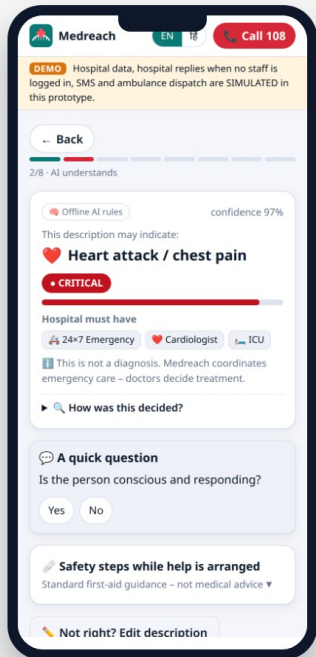


Emergency care · capacity · ambulance · handover

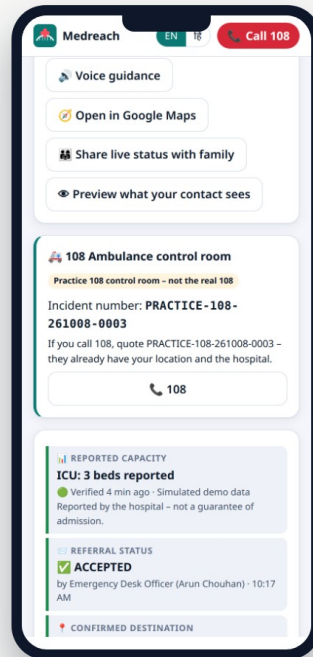


9:40 PM · Village near Berasia · Chest pain

A family knows they need help, but not which hospital can actually treat the case tonight. The nearest hospital is not automatically the right hospital.



AI flags a possible cardiac emergency



Hospital accepted + 108 incident number

0:00

Family calls / speaks in Hindi

No clear information about specialist or ICU.

0:30

MedReach identifies urgency indicators

Possible cardiac emergency; capability requirements are surfaced.

1:00

Hospitals ranked by capability + travel

Only suitable facilities remain in the shortlist.

2:00

Hospital accepts

ER bay / receiving team is prepared before travel.

2:30

Ambulance request created

Location, accepting hospital and incident details move together.

Result: the family leaves only after a capable hospital has said yes – and 108 already has the details.

THE PROBLEM

- Nearest hospital may not have the required specialist, ICU or equipment.
- Families often discover capacity problems only after reaching the gate.
- Hospital status can become outdated without a clear data owner.
- Rural users may have weak 2G / no reliable mobile data.

1,68,491 road-accident deaths in India in 2022 (MoRTH)

CORE ENGINE

**AI-assisted prioritization +
capability matching + live
capacity + routing + audit**

MEDREACH CONNECTS

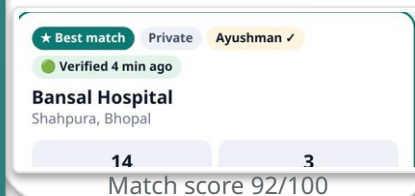
CITIZEN / FAMILY

Voice or text emergency input

Location + basic details

See matched hospitals

Track ambulance / status



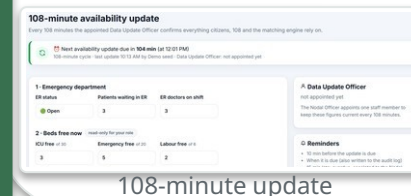
HOSPITAL

Receive pre-alert

Accept / decline referral

Reserve receiving capacity

Update beds, doctors, resources



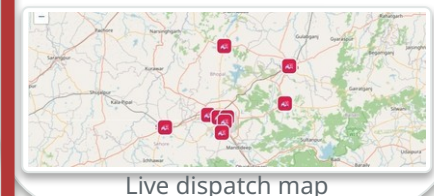
CONTROL ROOM

Incident visibility

Dispatch coordination

Ambulance status

Signed handover



The goal is not diagnosis. The goal is to reduce avoidable delay between the emergency, the right facility and the ambulance response.

SOLUTION PROPOSED

From first words to a prepared emergency room

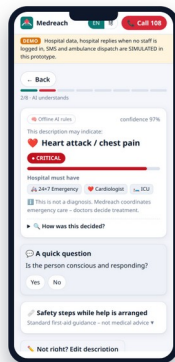
1 Speak / type

Voice, text or basic case details



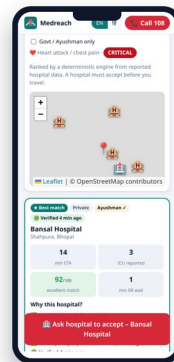
2 Prioritize

Red flags + urgency indicators



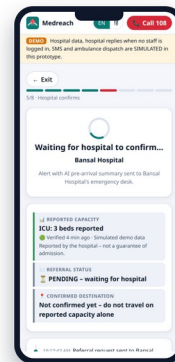
3 Match

Specialist, ICU, equipment, blood, distance



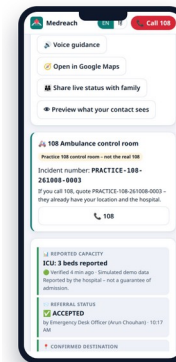
4 Accept

Hospital confirms capability



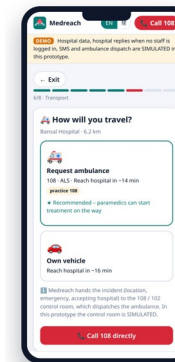
5 Reserve

Receiving bay / resource prepared



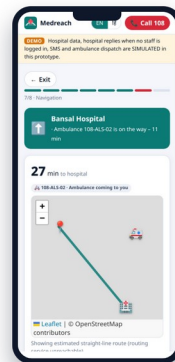
6 Dispatch

108 / 102 hand-off + tracking



7 Arrive

ER receives structured pre-alert



OPERATING RULES

Built into the product, not just policy.

ACCEPT FIRST

Hospital accepts before the family travels.

CAPABLE ONLY

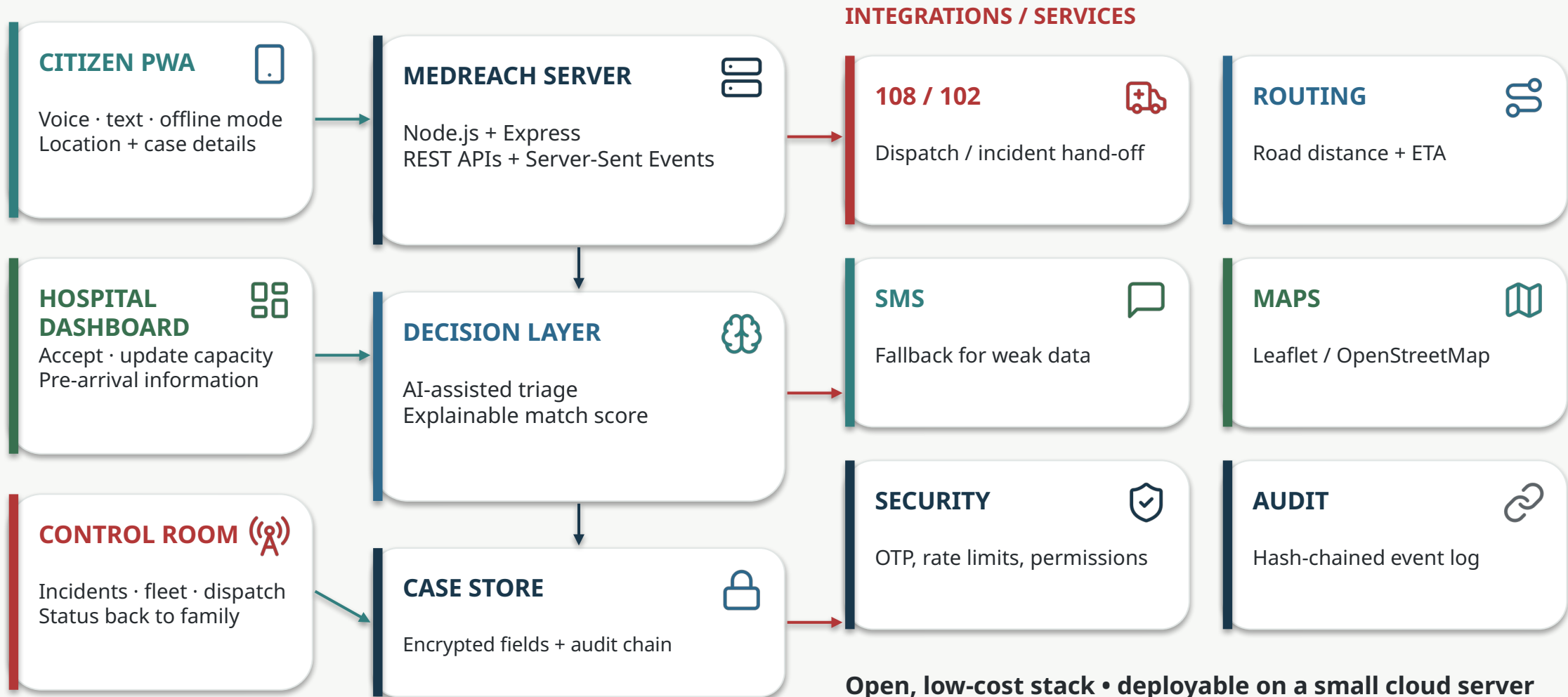
Only facilities that can handle the case are ranked.

NO DIAGNOSIS

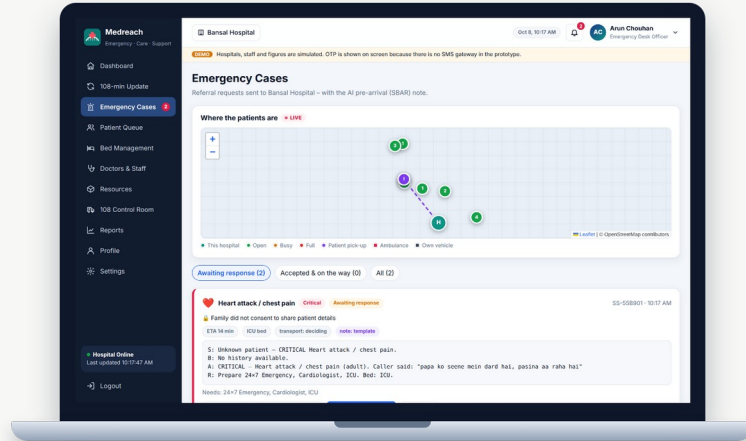
AI uses "may indicate" language; it does not diagnose.

108 ALWAYS

Call 108 remains visible and available.



Gemini 2.5 Flash · Web Speech · Tesseract.js · AES-256-GCM · HMAC



Hospital dashboard – incoming referral with AI pre-arrival note

16

hospitals · 4 districts

9

practice 108 ambulances

94

automated tests passing

4

network levels supported

WHAT THE MVP DEMONSTRATES

Citizen flow

Emergency input → matched facilities → acceptance status

Hospital flow

Incoming referral → capability/capacity update → accept

Control-room flow

Incident → ambulance assignment → status updates

Rural fallback

SMS code / low-bandwidth path when internet is unreliable

Handover

Structured pre-arrival information reaches the receiving team

Prototype today: 16 hospitals · 4 districts (sample data) · 9 practice ambulances · 94 automated tests

Capability	Calling around	Map apps	MedReach
Knows specialist / ICU availability	~ Usually no	✗ No	✓ Yes
Hospital confirms before travel	✗ No	✗ No	✓ Yes
Explainable match score	✗ No	✗ No	✓ Yes
Hindi / Hinglish input	~ Phone dependent	~ Limited	✓ Yes
2G / SMS / offline path	~ Variable	✗ No	✓ Yes
Pre-arrival hospital handover	✗ No	✗ No	✓ Yes
Ambulance + accepting hospital linked	~ Manual	✗ No	✓ Yes

Positioning: MedReach does not replace hospitals or 108/102. It connects the pieces and makes the hand-off explicit.

FIRST-TIME MATCH



Reduce unsuitable hospital visits

RIGHT FACILITY

PRE-ALERT



Hospital knows before arrival

READY ER

LIVE STATUS



Family sees progress

LESS UNCERTAINTY

RURAL ACCESS



2G / SMS / offline pathway

MORE REACH

CITIZENS & FAMILIES

- Speak instead of filling forms during panic.
- See whether a hospital has accepted.
- Track ambulance / incident status.
- Reduce avoidable travel between facilities.

HOSPITALS

- Receive a structured pre-arrival note.
- Control who updates capacity.
- Prepare receiving space before arrival.
- Maintain an auditable case trail.

HEALTH / CONTROL ROOM

- Single incident view.
- Connect location to accepting facility.
- Monitor ambulance movement.
- Use demand data for planning.



Bansal Hospital

Ambulance 108-ALS-02 is on the way – 11 min



Heart attack / chest pain Critical Awaiting response

Family did not consent to share patient details

ETA 14 min ICU bed transport: deciding note: template

S: Unknown patient – CRITICAL Heart attack / chest pain.
B: No history available.
A: CRITICAL – Heart attack / chest pain (adult). Caller said: "papa ko seene mein dard"
R: Prepare 24x7 Emergency, Cardiologist, ICU. Bed: ICU.



Scale path: pilot in a Bhopal-division cluster → state-wide hospital network → integration with any state 108/102 CAD.

WHY SHOULD WE SELECT YOU?

A practical, demonstrable and scalable approach

1 Works as an end-to-end workflow

Citizen → capable hospital → ambulance → tracking → handover.

2 Designed for real operating constraints

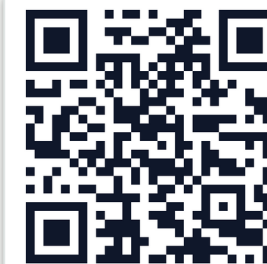
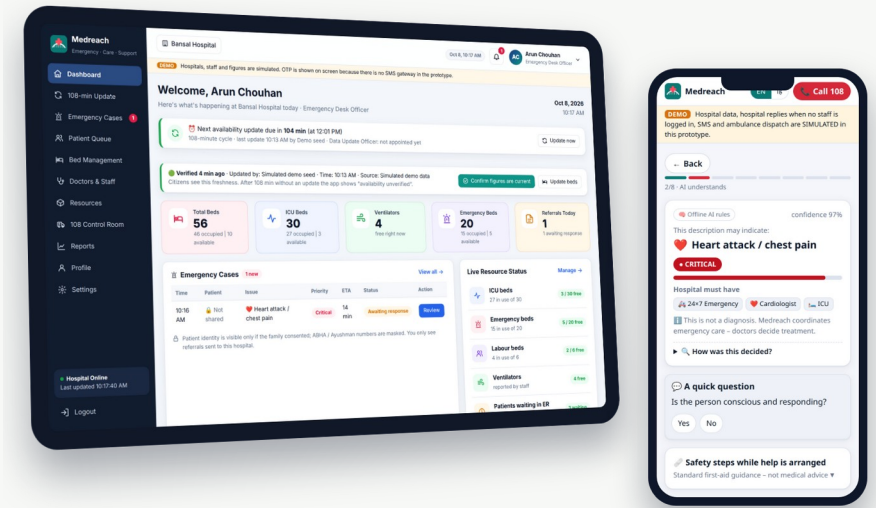
Hindi input, low bandwidth, SMS fallback and explicit hospital acceptance.

3 Responsible use of AI

AI assists prioritization and matching; it does not diagnose or replace clinicians.

4 Low-cost and integrable

Web-first architecture, open-source components and clear integration points.



SCAN TO TRY IT LIVE

medreach-2.onrender.com

Report in Hindi → hospital accepts →
108 dispatch → live tracking

LIVE PROTOTYPE

medreach-2.onrender.com | GitHub: prakharporwal518-collab/Medreach

MedReach connects the emergency journey — it does not replace emergency services.